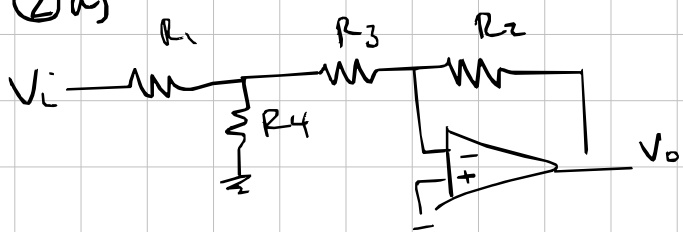
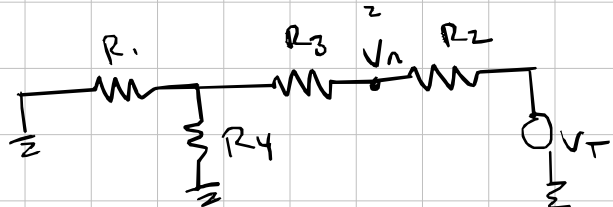


② a)



$$\frac{500}{1+jf/20} \text{ V/mV} \rightarrow a_o = 5 \times 10^5 \text{ V/V}$$

$$f_t = 10 \text{ MHz}$$



$$V_n = V_T \frac{R_3 + R_1 \parallel R_4}{R_3 + R_1 \parallel R_4 + R_2} = V_T \frac{R_3 (R_1 + R_4) + R_1 R_4}{(R_3 + R_2)(R_1 + R_4) + R_1 R_4}$$

$$\beta = -\frac{V_o}{V_T} = \frac{V_n}{V_T} = \frac{R_3 (R_1 + R_4) + R_1 R_4}{(R_3 + R_2)(R_1 + R_4) + R_1 R_4}$$

Let $R_3 = 5k$, $R_2 = 1M$, $R_1 = 5k$, $R_4 = 2.5k$

$$\beta = 0.00662 \rightarrow 1/\beta = 151$$

$$f_B = (1 + a_o \beta) f_g = 66.24 \text{ kHz}$$

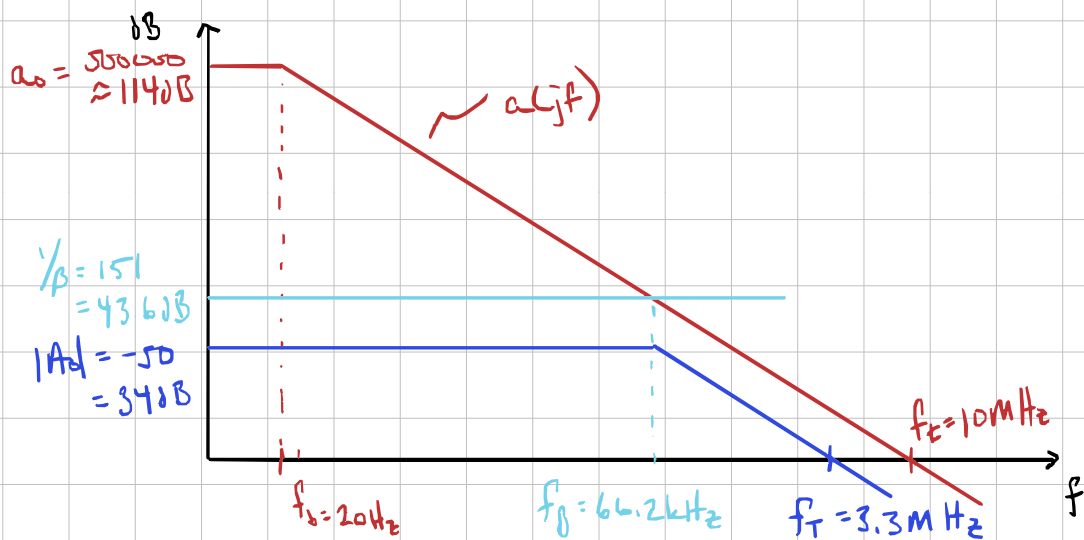
Approx: $f_B \approx \beta f_t = 66.22 \text{ kHz}$

$$A_{ideal} = -R_2 R_4 / (R_1 R_3 + R_1 R_4 + R_3 R_4) = -50$$

$A_o \approx A_{ideal}$ (a_o large $\rightarrow a_o \beta$ still large)

$$f_T \approx |A_o| / f_B = 3.3 \text{ MHz}$$

$$A_c(f) = -50 / (1 + jf / 66.24 \text{ kHz})$$



b) Recalculating β for $R_4 = 10k\Omega$:

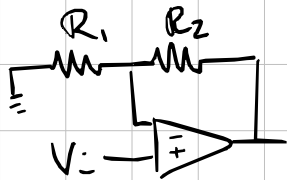
$$\beta = 0.00826, \quad f_B = \beta f_t = 82.6 \text{ kHz}$$

$$1/\beta = 121$$

for reference: $A_{ideal} = -80$! Gain went up, but so did bandwidth! This goes against the typical gain vs. tradeoff concern.

The reason: the change made caused f_T to approach f_t in addition to also increasing A_{ideal} . This would not have been possible if $f_T = f_t$ (like in non-inverting amplifier case).

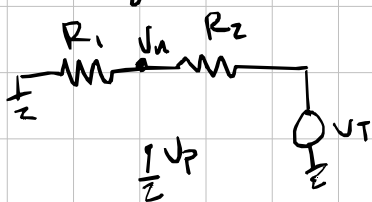
③ a)



$$a(jf) = 5 \times 10^5 / (1 + jf/20\text{Hz})$$

$$A_{ideal} = 100$$

(same analysis from lecture)



$$V_d = -V_n$$

$$\beta = -\frac{V_d}{V_T} = -V_n = \frac{R_1}{R_1 + R_2}$$

$$A_{ideal} = 1/\beta \rightarrow \beta = 0.01$$

$$f_B = f_L \beta = (5 \times 10^5)(20\text{Hz})(0.01) = 100\text{kHz}$$

$$A(jf) = 100 / (1 + jf/100\text{k})$$

$$f_B = 100\text{kHz}$$

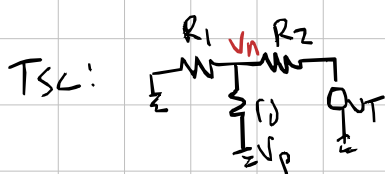
b) R_i (blackman's)

could be taken from previous lecture as well



let's assume $r_o \ll R_2 \dots$

$$r_o = r_o + R_1 \parallel R_2 = r_o + \frac{R_1 R_2}{R_1 + R_2}$$



$$V_n = \frac{R_1 \parallel r_o}{R_1 \parallel r_o + R_2} V_T = \frac{R_1 r_o}{R_1 r_o + R_2 (R_1 + r_o)} V_T$$

$$T_{SC} = a(jf)$$



$$V_n = V_p \rightarrow T_{OL} = 0$$

$$R_i = \left(r_o + \frac{R_1 R_2}{R_1 + R_2} \right) \left(1 + \frac{R_1 r_o}{R_1 r_o + R_2 (R_1 + r_o)} a(jf) \right)$$

if $r_o \gg R_1, R_2$, then

$$R_i \approx r_o \left(1 + \frac{R_1 r_o}{R_1 r_o + R_2 r_o} a(jf) \right) = r_o (1 + \beta a(jf))$$

$$R_i = r_o \left(1 + \frac{R_1}{R_1 + R_2} \frac{a_o}{1 + jf/f_B} \right)$$

$$= r_o \left(\frac{(R_1 + R_2)(1 + jf/f_B) + R_1 a_o}{(R_1 + R_2)(1 + jf/f_B)} \right)$$

$$= r_o \left[\frac{R_1 a_o}{R_1 + R_2} + 1 + jf/f_B \right] / (1 + jf/f_B)$$

$$= r_o \left(1 + \frac{R_1 a_o}{R_1 + R_2} \right) \left[\frac{1 + jf/f_B}{1 + jf/f_B} \right] \quad f_B = \left(1 + \frac{R_1 a_o}{R_1 + R_2} \right) f_o$$

$$= (1 + \beta a_o) f_o \approx \beta f_L$$

$$f \rightarrow 0: R_i = r_o \left(1 + \frac{R_1 a_o}{R_1 + R_2} \right) = 100\text{M}\Omega$$

$$100\text{M}\Omega = r_o (1 + \beta a_o) = r_o (1 + 0.01 \cdot 5 \times 10^5)$$

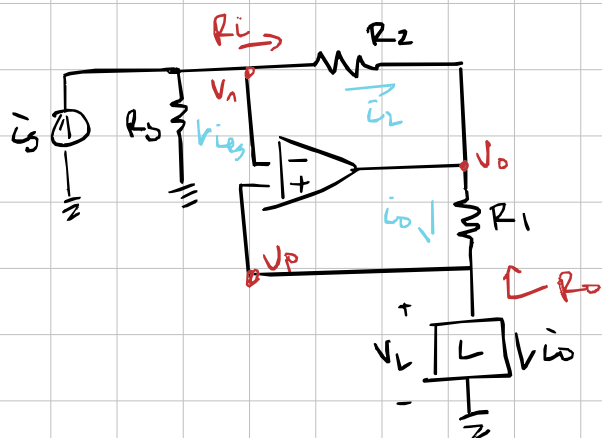
$$r_o = 20\text{k}\Omega \leftarrow \text{not realistic!}$$

$$f \rightarrow \infty: R_i = r_o (1 + \beta a_o) \frac{f_o}{f_B} = r_o (1 + \beta a_o) \frac{f_o}{f_o (1 + \beta a_o)} = r_o$$

$$R_i |_{f \rightarrow \infty} = 20\text{k}\Omega$$



④ Bask 6.25



$$R_1 = R_2 = 10 \text{ k}\Omega$$

$$f_t = 10 \text{ MHz} \quad r_s = \infty \quad r_o = 0$$

$$R_s = 30 \text{ k}\Omega \quad R_L = 2 \text{ k}\Omega$$

$$\text{KCL @ } V_n: \quad i_s = i_{R_s} + i_2 = \frac{V_n}{R_s} + \frac{V_n - V_o}{R_2} = V_n \left(\frac{1}{R_s} + \frac{1}{R_2} \right) - \frac{V_o}{R_2}$$

$$V_n = \left[i_s + \frac{V_o}{R_2} \right] \frac{R_s R_2}{R_s + R_2}$$

$$V_p = V_L, \quad V_o = a(jf) (V_p - V_n) = a(jf) \left(V_L - \left[i_s + \frac{V_o}{R_2} \right] \frac{R_s R_2}{R_s + R_2} \right)$$

$$V_o - V_L = i_o R_1 \rightarrow V_o = i_o R_1 + V_L$$

$$i_o R_1 + V_L = a(jf) \left(V_L - \left[i_s + \frac{i_o R_1 + V_L}{R_2} \right] \frac{R_s R_2}{R_s + R_2} \right)$$

Since R_L is known, can get rid of V_L : $V_L = R_L i_o$

$$i_o R_1 + i_o R_L = a(jf) \left(i_o R_L - \left[i_s + \frac{i_o R_1 + i_o R_L}{R_2} \right] \frac{R_s R_2}{R_s + R_2} \right)$$

$$i_o (R_1 + R_L) / a(jf) + \frac{i_o (R_1 + R_L)}{R_2} \frac{R_s R_2}{R_s + R_2} - i_o R_L = -i_s \frac{R_s R_2}{R_s + R_2}$$

$$\frac{i_o}{i_s} = \frac{-R_s R_2 / R_s + R_2}{\frac{R_s (R_1 + R_L)}{R_s + R_2} - R_L + \frac{R_1 + R_L}{a(jf)}}$$

inserting #3:

$$A = \frac{i_o}{i_s} = -7.5 \text{ k} / 7 \text{ k} + 12 \text{ k} / a(jf) = -1.07 / 1 + 1.71 / a(jf)$$

$$a(jf) = a_o / 1 + jf/f_b \quad \text{only know } f_t = 10 \text{ MHz} = a_o f_b$$

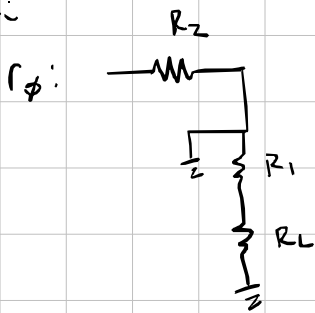
$$A = -1.07 / 1 + 1.71 (1 + jf/f_b) / a_o = -1.07 / 1 + \frac{1.71}{a_o} + j \frac{1.71}{a_o} \frac{f}{f_b}$$

$$a_o \text{ should be } \gg 1.71! \therefore \frac{1.71}{a_o} \ll 1$$

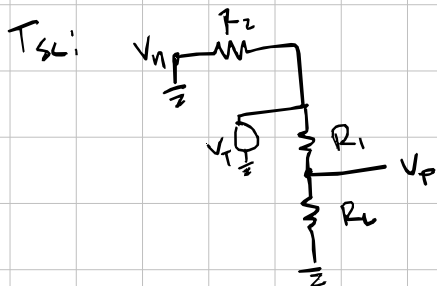
$$A \approx -1.07 / 1 + jf/f_b \quad f_b = f_b a_o / 1.71 = f_t / 1.71 = 5.83 \text{ MHz}$$



④ z_i



$$r_\phi = R_2 = 10k$$



$$V_d = V_p - V_n = V_p = \frac{R_L}{R_1 + R_L} V_T = 0.16 V_T$$

$$T_{SL} = -0.16 \text{ a}(jf)$$

T_{OL} : same as above except V_n floating $\rightarrow V_n = V_T$

$$V_d = V_p - V_n = 0.16 V_T - V_T = -0.83 V_T$$

$$T_{OL} = 0.83 \text{ a}(jf)$$

$$z_i = 10k \left(\frac{1 - 0.16 \text{ a}(jf)}{1 + 0.83 \text{ a}(jf)} \right) = 10k \left(\frac{1 + jf/f_b - 0.16 \text{ a}_o}{1 + jf/f_b + 0.83 \text{ a}_o} \right)$$

$$= 10k \left(\frac{-0.16 \text{ a}_o}{0.83 \text{ a}_o} \right) \left(\frac{1 - jf/f_z - 1/0.16 \text{ a}_o}{1 + jf/f_p + 1/0.83 \text{ a}_o} \right)$$

Both terms insignificant

$$z_i(jf) = 2k \left(\frac{1 - jf/f_z}{1 + jf/f_p} \right)$$

$$f_z = 0.16 \text{ a}_o f_b = 0.16 f_k$$

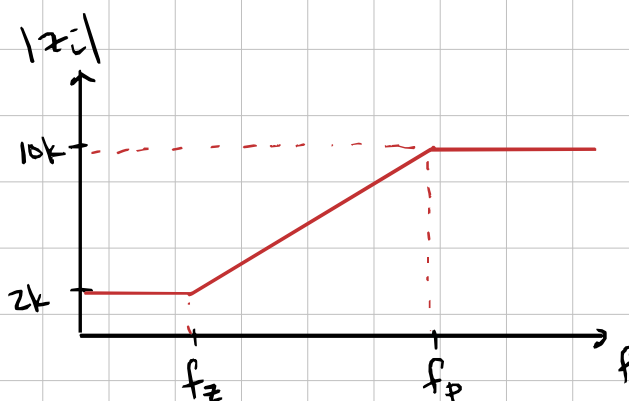
$$f_p = 0.83 \text{ a}_o f_b = 0.83 f_k$$

the zero with $-(1 - jf/f_z)$ means that it will have an opposite phase contribution as opposed to $(1 + jf/f_z)$; however, magnitude is the same.

Note that $f_z < f_p \rightarrow R_i$ increases with f .

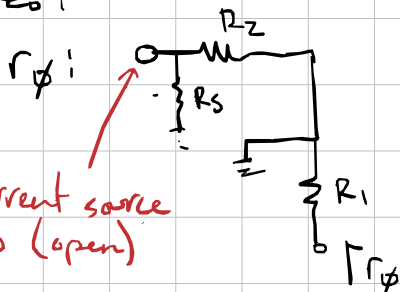
$$f \rightarrow 0, z_i = 2k$$

$$f \rightarrow \infty, z_i = 2k \left(-f_p/f_z \right) = 10k$$



④

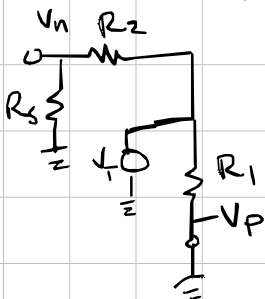
Z_o



current source
= 0 (open)

$$r_\phi = R_1 = 10k$$

T_{sc} :



$$V_o = V_p - V_n = -V_n$$

$$V_n = V_T \frac{R_s}{R_s + R_z} = 0.75 V_T$$

$$T_{sc} = 0.75 a(jf)$$

T_{oc} : Same as above except V_p floating $\rightarrow V_p = V_T$

$$V_o = V_T - 0.75 V_T = 0.25 V_T$$

$$T_{oc} = -0.25 a(jf)$$

$$Z_o = 10k \frac{1 + 0.75 a(jf)}{1 - 0.25 a(jf)} = 10k \frac{1 + jf/f_b + 0.75 a_o}{1 + jf/f_b - 0.25 a_o}$$

$$\approx 10k \left(\frac{0.75 a_o}{-0.25 a_o} \right) \left(\frac{1 + jf/f_z + 1/0.75 a_o}{1 - jf/f_p - 1/0.25 a_o} \right) \quad \text{insignificant}$$

$$Z_o(jf) = -30k \frac{1 + jf/f_z}{1 - jf/f_p} \quad \begin{matrix} f_z = 0.75 f_t = 7.5 \text{ MHz} \\ f_p = 0.25 f_t = 2.5 \text{ MHz} \end{matrix}$$

$$f \rightarrow 0 \quad Z_o = -30k$$

$$f \rightarrow \infty \quad Z_o = -30k \frac{f_z}{f_z} = -30 \frac{2.5}{7.5} = -10k$$

